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Enhancing Code Compliance of Non-Engineered and Pre-Engineered Buildings in Urban Nepal through Retrofitting Interventions in RC Construction

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Abstract:	Nepal's earthquake risk is intensified by rapid urbanization, substandard construction practices, limited preparedness, and vulnerable reinforced (RC) buildings. Many RC structures are either non-engineered or built using outdated Mandatory Rules of Thumb (MRT) (NBC 205:1994), termed "pre-engineered", which no longer meet current standards. The Gorkha earthquake exposed severe vulnerabilities, highlighting the need for retrofitting interventions to enhance seismic resilience. While previous studies assessed existing low- to mid-rise RC buildings, significant gaps remain in the application of conventional methods, open-source software, and cost-effective retrofitting schemes that incorporate both traditional and innovative techniques across diverse building typologies. This study selected six typical buildings including symmetrical, plan-irregular, and vertically irregular structures of three- and four-stories. The seismic vulnerability of non-engineered and pre-engineered buildings within these typologies was assessed through non-linear static and dynamic analyses performed in OpenSees. Retrofitting methods, including RC jacketing, Steel jacketing, and Fiber-reinforced cementitious matrix (FRCM) including Glass FRCM and Hemp FRCM, were applied to upgrade these structures to meet the latest seismic design code NBC 105:2020. Incremental dynamic analysis (IDA) was conducted, and fragility functions were derived for each building model before and after retrofitting. The findings indicated that retrofitted models exhibited higher ductility and lateral load-carrying

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Abstract

Nepal's earthquake risk is intensified by rapid urbanization, substandard construction practices, limited preparedness, and vulnerable reinforced (RC) buildings. Many RC structures are either non-engineered or built using outdated Mandatory Rules of Thumb (MRT) (NBC 205:1994), termed "pre-engineered", which no longer meet current standards. The Gorkha earthquake exposed severe vulnerabilities, highlighting the need for retrofitting interventions to enhance seismic resilience. While previous studies assessed existing low- to mid-rise RC buildings, significant gaps remain in the application of conventional methods, open-source software, and cost-effective retrofitting schemes that incorporate both traditional and innovative techniques across diverse building typologies. This study selected six typical buildings including symmetrical, plan-irregular, and vertically irregular structures of three- and four-stories. The seismic vulnerability of non-engineered and pre-engineered buildings within these typologies was assessed through non-linear static and dynamic analyses performed in OpenSees. Retrofitting methods, including RC jacketing, Steel jacketing, and Fiber-reinforced cementitious matrix (FRCM) including Glass FRCM and Hemp FRCM, were applied to upgrade these structures to meet the latest seismic design code NBC 105:2020. Incremental dynamic analysis (IDA) was conducted, and fragility functions were derived for each building model before and after retrofitting. The findings indicated that retrofitted models exhibited higher ductility and lateral load-carrying capacity, reduced inter-story drifts, and delayed damage onset under increasing seismic loads, thereby meeting current code requirements. At 0.4 PGA, the probability of exceeding the life-safety limit state decreased from 100% to 45% in non-engineered buildings and from 90% to 38% in pre-engineered buildings. Glass-FRCM was the most effective, while Hemp-FRCM was least effective. Irregular structures required more extensive retrofitting than symmetrical ones.

Keywords:

Seismic vulnerability, Reinforced concrete buildings, Retrofitting, Nonlinear analysis, Nepal, OpenSees, Fragility curves.

1 Introduction

Nepal is a seismically active region where the Indian tectonic plate subducts under the Tibetan plate; the seismic history of Nepal shows the occurrence of large devastating earthquakes every 60-70 years [1]. This earthquake risk is further aggrandized into human disasters due to rapid urbanization, poor construction practices, and lack of disaster preparedness, especially in the urban neighborhoods of Nepal [2]. The history of building construction practice in Nepal dates back several thousand years, with the traditional settlements of the Kathmandu valley dating back to the thirteenth century. The current building stock of Nepal is heavily covered by RC construction, which began in the late 1970s. In particular in the last 3-4 decades, RC buildings have been heavily replacing adobe, stone, and brick masonry constructions. The 2021 census [3] presented a 22.4% proportion of RC buildings in the country, with 30% of them located in the Kathmandu valley [4]; a sharp rise from a staggering 10% from the 2011 census [5]. Nepal developed its first National Building Code (NBC) only in 1994 which came into legal effect later in 2006. This was NBC 205:1994 [6], also known as Mandatory Rules of Thumb (MRT) [7], which aimed to prescribe ready-to-use seismic design guidelines, including size of structural members, reinforcement detailing, and concrete grade. Consequently, many of these structures are either owner built or constructed by less experienced contractors on the basis of by-laws or the precursory MRT making them less resilient to strong motions. A more revised version of NBC 105 was released only recently in 2020 [8] after the Gorkha earthquake following the revised NBC 105:1994 that takes into account Nepal’s unique geography, probabilistic seismic hazard analysis and non-linear analysis.

The aftermath of Gorkha earthquake evidently highlighted the seismic vulnerabilities in the RC construction of Nepal. Approximately, 80% of these structures experienced various degrees of damage, with about 70% falling into the categories of non-engineered and pre-engineered buildings [9]. Owner-built houses were completely destroyed, while MRT-based buildings suffered partial damage given the inherent code deficiencies [10]. Common structural deficiencies in Nepalese RC buildings include the absence of ties in beam-column joints, inadequate confinement and lap lengths, low concrete strength, improper tie anchoring, irregular load distribution, and soft-story and short-column effects [11]. More so, even many engineered buildings failed to meet seismic standards due to secondary issues like design flaws, poor site workmanship, and subpar construction practices. Hence, the overall vulnerability of the reinforced concrete (RC) building stock in Nepal underscores the need for retrofitting interventions in the vast number of existing RC structures to enhance the country’s seismic resilience.

Focusing on the impact of 2015 Gorkha earthquake, Gautam et al. [2] highlighted the need for urgent improvement in building codes and strengthening techniques. Chaulagain et al. [10] and Gautam et al. [9] highlighted the seismic vulnerability of non-engineered RC buildings in Nepal, emphasizing the need for retrofitting and strengthening frameworks to address deficiencies and improve resilience against future earthquakes. Furthermore, Chaulagain et al. [10] assessed the seismic strengthening solutions for existing low-rise RC buildings in Nepal, exploring RC shear walls, steel bracing, and RC jacketing, demonstrating that these methods significantly enhance seismic safety. Sthapit et al. [12] documented the retrofitting of a severely damaged RC frame building in Kathmandu Valley, focusing on immediate preventive measures, analytical assessments, and phased construction stages to enhance the building’s seismic resilience and structural integrity. Pandit et al. [13] analyzed the seismic performance of reinforced concrete structures in Nepal before and after the 2015 earthquake, with a focus on the ‘R’ value in safety assessments and examining shifts in design and construction practices. Their study provided insights into the over-strength, ductility, and nonlinear behavior of selected buildings to better understand the impact of the earthquake on seismic design approaches in Nepal.

Gautam et al. [11] focused on the seismic vulnerability assessment of reinforced concrete buildings in Nepal, discussing the importance of studying non-structural elements and the use of advanced analytical and experimental tools for seismic risk mitigation. They also presented fragility functions for different building components, emphasizing the need for additional data to reduce uncertainties and aiming to compare their findings with existing literature to enhance the reliability of seismic fragility modeling in Nepal. Adhikari et al. [14] discussed the seismic vulnerability and retrofitting schemes for low-to-medium-rise reinforced concrete buildings in Nepal. Their study analyzed the structural performance and seismic response of different building models using various analysis techniques, such as static pushover analysis, adaptive pushover analysis, and dynamic time history analysis.

Dumaru et al. [15] performed a comprehensive cost-benefit analysis on retrofitting both non-engineered and engineered buildings in Nepal, using a probabilistic framework to assess the effectiveness of various retrofit techniques post-earthquake. Their study aimed on assessing the economic feasibility of retrofitting by comparing expected annual losses, life cycle losses, and sensitivity analyses across different building

types. Similarly, Badal et al. [16] examined the seismic performance of representative building models in Kathmandu Valley, highlighting the impact of earthquakes on these structures and the pressing need to retrofit older buildings to improve safety.

Bhusal et al. [17] assessed the seismic performance of various prototype of RC beam-column joints designed in accordance to NBC 205:1994, NBC 205:2012, and the revised NBC 105:2020. They investigated retrofit strategies aimed at enhancing both strength and ductility. In a subsequent study, Bhusal et al. [18] examined the seismic performance of low-rise RC buildings in Nepal, again comparing designs based on three codes. Their findings underscore the efficacy of the NBC 105:2020 provisions in mitigating seismic losses. Their study focused on retrofitting RC columns using concrete, steel jacketing, and carbon fiber-reinforced polymer (CFRP) to enhance lateral strength and energy dissipation, while also employing machine learning to correlate peak ground acceleration with structural drifts for more practical applications. Gkournelos et al. [19] discussed the gaps in seismic retrofitting techniques and emphasized the importance of integrating modern seismic upgrading methods with energy efficiency measures. Ebead et al. [20] investigated the effectiveness of carbon and PBO Fabric-Reinforced Cementitious Matrix (FRCM) systems in improving the flexural capacity and deformation characteristics of RC beams. Through experimental tests on 12 RC beams with varying reinforcement ratios and FRCM strengthening systems, their study provides critical insights into the potential of FRCM as a viable technique for improving the performance of concrete structures. Gupta et al. [21] developed a numerical model for retrofitting non-engineered buildings using vegetal FRCM composites. Their findings indicate that 2-ply configurations of hemp and flax FRCM can significantly improve the base shear strength and ductility of the buildings, effectively reducing seismic vulnerability.

The research by the aforementioned authors on the seismic performance of existing low- to mid-rise RC buildings in Nepal, mostly in the Kathmandu Valley, helps identify the fragility, expected performance, and limitations of different categories of buildings. However, none of them have performed a comprehensive retrofitting analysis including both conventional and innovative retrofitting techniques. None of the studies mentioned above utilized distributed plasticity models, thus, there is a need for research employing conventional methods, comprehensive retrofitting schemes, and a diverse typology of both regular and irregular structures.

In this study, numerical analyses of six RC buildings, located in Lalitpur Metropolitan City (LMC), is conducted. These buildings include dominant typologies, namely symmetrical, plan-irregular, and vertically irregular structures of three- and four-story each. Non-linear static and dynamic analyses were employed to assess the current vulnerability of non-engineered and pre-engineered form of these buildings. Retrofitting interventions were applied to each building, including RC jacketing, steel jacketing, Glass-FRCM jacketing, and Hemp-FRCM jacketing. The effectiveness of each intervention was assessed through pushover analysis, time history analysis and subsequent fragility analysis.

2 Methodology

The floor plans collected from Lalitpur Metropolitan City (LMC) were analyzed to identify dominant typologies in the city that are representative of the RC buildings in the Kathmandu Valley. Most of the buildings could be subsumed under three main typologies: symmetrical/regular, irregular in plan, and vertically irregular. The selected buildings have two bays in one direction and three bays in the other with floor areas of less than 80 m², consistent with the finding by Chaulagain et al. [22] that most RC residential buildings in Kathmandu range between 65 m² and 90 m² as shown in Table 1. Residential buildings in the region are predominantly two to six stories, with the majority being three stories tall [22]. Therefore, three- and four-story buildings were taken for each of the considered typologies as shown in Figure 1. All of these building configurations are typical of urban areas in Nepal, including the Kathmandu Valley. Thus, the six selected representative buildings were structurally designed following three different design practices/guidelines: Non-Engineered Design, Pre-Engineered Design, and Engineered Design.

Non-engineered buildings refer to structures with minimal or substandard reinforcement detailing, typically built before the first seismic guidelines NBC 105:1994 and NBC 205:1994 and without adherence to any seismic standards. Pre-engineered buildings are designed based on the guideline of the NBC 205:1994, while engineered buildings are designed based on the latest seismic requirements specified in NBC 105:2020.

Among the floor plans collected, pre-engineered and engineered buildings had detailed structural plans, whereas non-engineered buildings lacked such details. To obtain the structural details of non-engineered building, a thorough literature review was conducted. Based on visual observations of RC

buildings that collapsed in Gorkha earthquake, Chaulagain et.al [22] provided detailed specifications of such owner-built non-engineered buildings. These buildings typically have column cross sections of 230 mm $\times$ 230 mm and beams with a rectangular cross section of 230 mm $\times$ 325 mm. The reinforced concrete floor slabs were 115 mm thick and the concrete and steel grades for all structural elements were 20 MPa and 415 N/m², respectively. All periphery walls and walls adjacent to stairs were 230 mm thick, while other internal partition walls were 115 mm thick (half-brick thick). Among nine of the owner-built building samples in the study by Adhikari et al.[14], four of them featured similar designs. Thus, for the non-engineered form, similar structural details were adopted based on these findings.

The structural details of the pre-engineered building was obtained in accordance with NBC 205:1994 [23], which provides well-documented Mandatory Rules of Thumb (MRT) guidelines. Similarly, the structural detailing of the engineered building was developed through a comprehensive design process using ETABS (Extended Three-Dimensional Analysis of Building Systems), a finite element analysis (FEA) software widely utilized for designing building structures. ETABS was specifically chosen for this study due to its extensive use in the engineering community to design RC buildings. This design adheres to all relevant codal provisions outlined in NBC 105:2020. As all typologies comply with the clauses of NBC 105 [24], the Equivalent Static Method (ESM) was employed for the engineered design of these buildings.

Material properties and loading conditions for all six buildings are assumed to be identical. Gravity loads were obtained from IS 875 [25] and seismic loads from NBC 105:2020, considering a residential building. Gravity loads include dead loads, live loads, and floor finishes, with dead loads encompassing the self-weight of beams, columns, slabs, and walls. The slab thickness is 125 mm, with internal and external wall thicknesses of 115 mm and 230 mm respectively. Wall load intensity is based on brick density (20 kN/m³ including plaster), with a 30% reduction in intensity for the walls with openings. The concrete considered is of normal-weight (25 kN/m³) with a compressive strength of 20 MPa, and steel reinforcements have a yield strength of 415 MPa and an elasticity modulus of 210 GPa. Live loads are 2 kN/m² for all slabs, except for roofs, which are 1.5 kN/m². During an earthquake, only 25% of these live loads are considered, as recommended by NBC 105.

The seismic design is carried out assuming ordinary structures (Importance factor=1) in Kathmandu, with a zone factor of 0.35 on medium soil type. The fundamental period of vibration is calculated using the empirical equation in NBC 105:2020. The ductility factors for ultimate limit state(ULS) and serviceability limit state(SLS) are 4 and 1 respectively, with over-strength factors of 1.5 and 1.25, corresponding to an RC-MRF system. A heuristic approach was employed to address critical stresses arising in sections at the ultimate limit state, such as axial loads in the columns and flexural loads in slabs and beams. Based on preliminary design and ETABS analysis, final dimensions for structural elements were established. All elements were designed and checked according to Indian Standards (IS) provisions. Beam sections with reinforcements at supports and mid-span are detailed in Table 3, and column longitudinal reinforcement is detailed in Table 2. All models in this study have a storey height of 3 m. In every storey, the beam has dimension of 230 $\times$ 350 mm.

Table 1: General description of building models.

Model	Plan Area (m ²)	No. of storey	Building Classification	Storey Height
1	78.1	3	Symmetric	3 m
2	60.84	3	Plan Irregularity	
3	77.01	3	Vertical Irregularity	
4	60.84	4	Symmetric	
5	57.83	4	Plan Irregularity	
6	60.84	4	Vertical Irregularity	

2.1 Non-linear Modeling

In non-linear analysis, geometric non-linearity and material non-linearity are commonly encountered. Material nonlinearity was the primary focus of this study, as it governs the behavior of structural components under inelastic deformations. Geometric nonlinearity is negligible for low-rise buildings due to

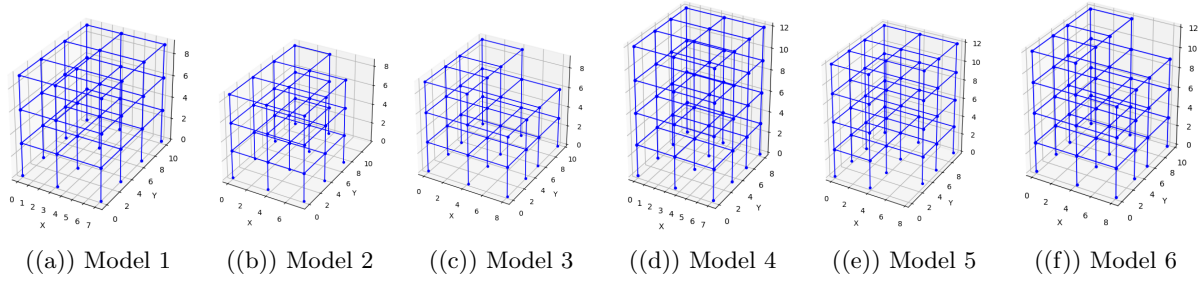


Figure 1: OpenSees Model of selected buildings

Table 2: Column sizes and reinforcement details for different building typologies.

Storey	Position	Non-Engineered	Pre-engineered	Code-Compliant
1st Floor	Corner	230 X 230, 4-16mm	270 X 270, 4-16mm	355x355, 8-20mm
	Face	230 X 230, 4-16mm	270 X 270, 4-16mm	355x355, 4-16mm+4-20mm
	Interior	230 X 230, 4-16mm	270 X 270, 4-16mm	355x355, 4-16mm+4-20mm
2nd Floor	Corner	230 X 230, 4-12mm	270 X 270, 4-16mm	355x355, 8-16mm
	Face	230 X 230, 4-12mm	270 X 270, 4-16mm	355x355, 8-16mm
	Interior	230 X 230, 4-12mm	270 X 270, 4-12mm	355x355, 4-16mm+4-12mm
3rd Floor	Corner	230 X 230, 4-12mm	230 X 230, 4-12mm	355x355, 8-16mm
	Face	230 X 230, 4-12mm	230 X 230, 4-12mm	355x355, 8-16mm
	Interior	230 X 230, 4-12mm	230 X 230, 4-12mm	355x355, 4-16mm+4-12mm

Table 3: Beam configurations for non-engineered and pre-engineered

Beam	Beam Configuration	
1st Floor	Top: 3-16mm, Bottom: 3-12mm	Top: 2-16mm, 1-12mm, Bottom: 2-16mm, 1-12mm
2nd Floor	Top: 3-16mm, Bottom: 3-12mm	Top: 2-12mm, 1-16mm, Bottom: 3#12
3rd Floor	Top: 2-16mm, Bottom: 2-12mm	Top: 2-12mm, 1-16mm, Bottom: 2-12mm

Table 4: Minimum concrete strength requirements for different building typologies

Typologies	Non-Engineered	Pre-engineered	Code-Compliant
Minimum Concrete Strength	15 MPa	15 MPa	20 MPa

their limited height and corresponding stability under lateral loads. The masses were lumped at the nodes, a standard practice to simplify the model while capturing essential dynamic effects [26].

Non-linearity in structures can be modeled through either Distributed Plasticity (DP) approach or Concentrated Plasticity (CP) approach [27]. The CP approach localizes nonlinearity at specific points, typically at the ends of elements, while the rest of the element is treated as linear element. In contrast, the DP approach allows nonlinearity to occur throughout the entire element, assigning integration points along the length of the element to capture the spread of plasticity. The CP model is simpler to implement and computationally faster but is limited by the plastic hinge length, restricting its applicability. Conversely, DP model allows plasticity to propagate along the element, enabling flexibility to yield at any point along its length. This study adopted the DP approach, modeling beams and columns as nonlinear elements using a ‘force-based beam-column element’ [28] within the OpenSees framework [29], with each nonlinear fiber assigned at five integration points.

The material models selected for this analysis were ‘Concrete02’ for concrete and ‘Steel02’ for reinforcing steel, as shown in Figure 2 and Figure 3. These models are defined using the ‘uniaxialMaterial’ command, representing uniaxial force-deformation relationships. The ‘Concrete02’ model is based on the modified Kent and Park model [30]. The ‘Steel02’ model, which follows the Giuffre-Menegotto-Pinto relationship, includes strain hardening and was parameterized according to established guidelines [31]. Given that pre-engineered and engineered buildings are well-confined by transverse reinforcements, core concrete was modeled with increased compressive strength following Mander’s Confinement model. So, the core concrete was divided into a grid of fibres, with more fibres used in the confined core region to capture the enhanced compressive strength. Developed by Mander et al. [32], this model effectively

predicts the stress-strain response of confined concrete under both monotonic and cyclic loading, making it particularly useful for simulating seismic behavior. The stress and strain diagram of confined and unconfined concrete is shown in Figure 4.

In the fiber-based modeling approach, the cross-section of a structural member are assigned to specific locations called integration points (IPs) within a beam-column element. Each fiber is assigned its constitutive relations to capture non-linearity. These IPs discretize the element and determine displacements, with their number and placement affecting accuracy and computational efficiency. The fiber cross-sections are applied along the entire element length, though they can also be used for a plastic zone where inelastic behavior is expected.

The “nonlinearBeamColumn” element was used for beams and columns, employing five IPs as commonly used in the literature including Harrington et al. [33, 34]. It considers the spread of plasticity along the element and integrates using the Gauss-Lobatto quadrature rule with two integration points at the ends and others at regular intervals along the element.

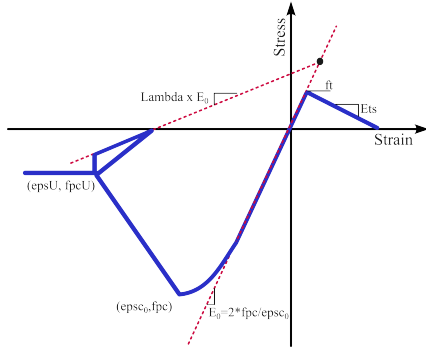


Figure 2: Hysteretic behavior of ‘Concrete02’

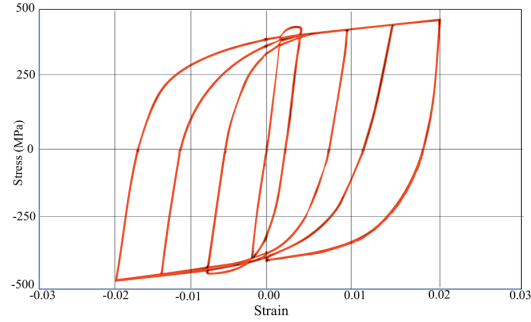


Figure 3: Hysteretic behavior of ‘Steel02’

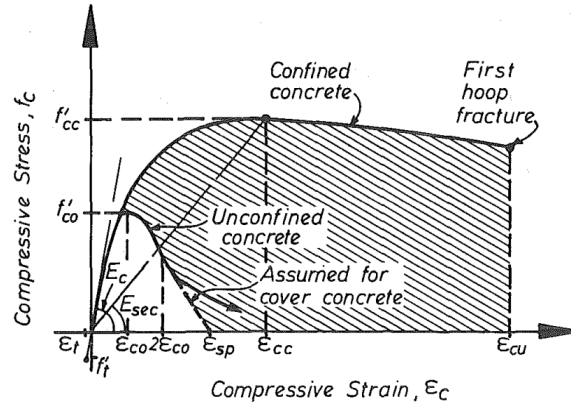


Figure 4: Stress-strain diagram for confined and unconfined concrete
[Source: Mander et al. [32]]

2.2 Pushover Analysis

Non-linear static or pushover analysis is a performance-based evaluation method used to understand the behaviour of structures under increasing lateral loads. Unlike linear analysis, pushover analysis accounts for material and geometric non-linearity and helps to identify weak points in buildings and their response to seismic events [35]. The curve generally represents the first mode response for buildings with fundamental periods of vibration up to about one second. For more flexible buildings with longer periods, higher mode effects should be considered [36]. The method provides crucial insights into the ultimate strength, deformation capacity, and failure mechanisms of structures, making it an essential methodology to assess and retrofit buildings that may not meet current seismic codes [37].

In this study, pushover analysis was performed using OpenSees to evaluate the seismic performance of six building typologies. Following the procedure specified by NBC 105:2020, a gravity analysis was

Table 5: Material properties for concrete and steel

Parameters	Value	Units
Yield strength of steel (f_y)	415	MPa
Ultimate strength of steel (f_u)	485	MPa
Yield strain of steel (ε_y)	0.002075	–
Ultimate strain of steel (ε_u)	0.144	–
Modulus of elasticity of steel (E_s)	2×10^5	MPa
Modulus of elasticity of concrete	$5000\sqrt{f_{ck}}$	MPa
Poisson's ratio for concrete (ν)	0.2	–
Shear modulus ($G = E/2(1 + \nu)$)	9316.95	MPa
Unit weight of concrete	25	kN/m ³
Concrete strain at maximum stress in compression	0.002	–
Concrete strain at ultimate stress in compression	0.02	–

performed to account for the structure's self-weight, with the resulting displacements and reactions from the final step applied to the first step of the pushover analysis. Lateral loads proportional to the first mode shape and in the weaker direction were applied at the master node of each floor, incrementally increasing until the specified displacement was reached. During the analysis, displacements at the roof's center of gravity and base reactions were recorded at each step. Base reactions were summed to calculate base shear, which was then plotted against displacement to create a pushover curve, which represents the overall structural response as the building approaches dynamic instability during a significant earthquake [38]. This curve was used to assess the seismic capacity of non-, pre-, and engineered buildings. For buildings with deficiencies, retrofitting measures were introduced to improve their seismic response. The extent of retrofitting was determined by whether the retrofitted structure achieved the peak base shear of an engineered building.

2.3 Dynamic Time History Analysis

Non-linear dynamic analysis is a robust technique for evaluating the dynamic response of structures subjected to seismic forces. Unlike static approaches, it directly solves the differential equations of motion to account for the time-varying effects of ground acceleration [14]. This method offers a more detailed understanding of structural behavior during real seismic events, capturing post-elastic performance and potential failure mechanisms. It is particularly beneficial for complex structures, where static analyses, such as pushover, may not accurately predict dynamic behaviors like torsion, damping, and P-Delta effects. This analysis also helps characterize the probabilistic relationships between seismic intensity measures (IMs) and engineering demand parameters (EDPs) [33]. In this study, the relevant IM is Peak Ground Acceleration (PGA), while the EDP is the peak inter-story drift ratio.

2.3.1 Selection of Ground Motion

The selection of ground motion plays a critical role in developing fragility curves through nonlinear time history analysis, as ground motion parameters significantly impact the seismic performance of structures. Achieving the optimal number of ground motions requires a balance between computational efficiency and the desired level of accuracy. In this study, 11 real earthquake ground motion records from the PEER database were utilized. In NBC 105: 2020, Nepal's seismic hazard is determined through probabilistic seismic hazard assessment (PSHA), with a peak ground acceleration (PGA) of 0.35g in Kathmandu valley for a 475-year return period. Given the valley's loose and alluvial soil deposits, it is essential to select earthquake ground motion records that reflect the local geological conditions to ensure accurate predictions of structural responses.

However, the ground motion records listed in Table 6 correspond to various geological conditions around the world. These records were adjusted to align with the target soil type A design spectrum of NBC 105 by using wavelet-based modification techniques in SeismoMatch [39], within the required period range for the building models analyzed. Response spectrum plots, after adjustment, are presented in Figure 5, showing the comparison between the modified records, the target spectrum, and their mean values. The selected ground motions have magnitudes ranging from approximately 6 to 8, encompassing strike-slip and reverse fault types and representing near-field earthquakes.

Table 6: Summary of selected ground motion with their details

SN	Earthquake	Year (AD)	Magnitude (Mw)	PGA (g)	Hypocenter (km)
1	Big Bear	1992	6.3	0.25	320
2	ChiChi	1999	7.6	0.9	19
3	Imperial Valley	1979	6.5	0.2	32
4	Kern	1952	7.3	0.5	20
5	Landers	1992	7.3	0.8	32
6	Loma Prieta	1989	6.9	0.4	10
7	Northridge	1994	6.7	0.3	24
8	Palm	1927	6.9	0.35	320
9	Parkfield	2004	6	0.2	24
10	San Fernando	1971	6.6	0.3	24
11	Whittier	1987	6.1	0.25	48

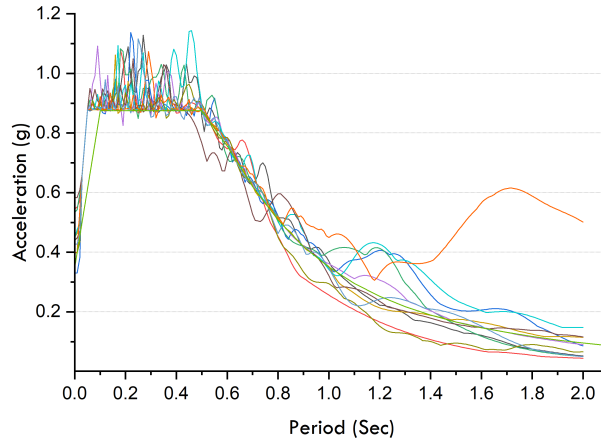


Figure 5: Spectral Matching of Selected Ground Motion

2.3.2 Incremental Dynamic Analysis

Incremental Dynamic Analysis (IDA) is a method that involves performing multiple nonlinear time history analyses on a structural model using a suite of ground motion records, each scaled to different intensity levels [40]. IDA is regarded as the dynamic equivalent of static pushover analysis and is utilized to evaluate both the demand and capacity of structures [40]. Thus, IDA is performed to generate curves of the EDP against the IM for all the applied ground motions. These curves can subsequently be used to develop fragility curves for the buildings. With the selected ground motions, IDA is performed in OpenSees while tracking the inter-story drift ratio, and a set of different values of peak inter-story drift ratio for different PGAs are obtained. The analysis followed the guidelines of NBC 105:2020, including three-dimensional modeling of the structure, accounting for material degradation and stiffness degradation. Gravity loads and P-Delta effects were also incorporated into the model. Dynamic responses, such as drift, were recorded at each increment, and response plots were generated to show the relationship between structural behavior and ground motion intensity. The final step involved the derivation of fragility curves for the structure, which were used to assess vulnerabilities and recommend potential retrofitting measures.

3 Retrofitting Interventions

Retrofitting interventions encompass any technique used to enhance the strength, stiffness, and ductility of structural components or the building as a whole [10]. The choice of an appropriate retrofitting method is influenced by several key factors, including: adherence to building codes regarding design, materials, and construction; ensuring the solution aligns with the functional requirements of the structure; consideration of both direct and indirect costs, such as relocation expenses; the availability of required construction techniques, materials, and skilled labor; improving the building's safety post-intervention;

and maintaining the aesthetic appearance of the structure.

There are two primary methods to enhance the seismic capacity of existing structures: global-level and local-level interventions. Global interventions involve modifications to the entire structural system, while local interventions focus on enhancing the ductility of specific vulnerable components to meet required limit states. This study focuses exclusively on local-level interventions, targeting only the vulnerable components without altering the building’s architectural design. Local interventions are cost-effective and are commonly employed to retrofit non-ductile elements, improving their strength, stiffness, and ductility.

Four local retrofitting options were examined in this study: Reinforced Concrete jacketing, steel jacketing, Glass FRCM (Fabric Reinforced Cementitious Matrix) jacketing, and vegetal hemp FRCM jacketing. These techniques were applied to the columns of both non-engineered and pre-engineered buildings, resulting in the retrofitting of 48 structures. The goal was to enhance the seismic performance of these buildings, ensuring that the pushover curves of the retrofitted structures align with those of engineered buildings. The retrofitted structural systems were then analyzed using the same evaluation procedures to verify their improved performance.

3.1 Reinforced Concrete Jacketing

Reinforced concrete (RC) jacketing is a widely recognized technique for strengthening existing structural elements to address deficiencies in strength and deformation capacity. This technique is particularly effective for improving the development of longitudinal reinforcement and addressing issues with lap splices, as identified during detailed structural evaluations. Experimental studies have consistently demonstrated the effectiveness of RC jacketing in restoring and enhancing the performance of structural elements, particularly columns [34]. RC jacketing significantly enhances the shear capacity, flexural strength, and ductility of damaged or underperforming members [41, 42]. However, while column jacketing increases the overall stiffness of the building, it also leads to an increase in seismic forces, which may result in higher displacements and lateral forces acting on the structure [43]. Thus, a change in stiffness does not directly translate to reductions in drift but affects the overall lateral behavior of the structure.

In this study, RC jacketing was applied to columns in accordance with IS 15988:2013 (Seismic Evaluation and Strengthening of Existing Reinforced Concrete Buildings) [44]. The jacketing procedure was performed on all four sides of the columns. The strength of the new concrete used in the jacketing was designed to be at least 5 MPa greater than that of the existing concrete in the columns. The design of the additional concrete thickness and the reinforcement requirements was based on the following calculations:

$$A_c = \left(\frac{3}{2}\right) A'_c \quad \text{and} \quad A_s = \left(\frac{4}{3}\right) A'_s$$

where:

- A_c and A_s represent the actual amount of concrete and steel to be provided in the jacket.
- A'_c and A'_s represent the concrete and steel values after deducting the contributions of the existing concrete and steel from the required amounts.

The jacketing was applied to all columns within the building, ensuring a uniform enhancement of the structure’s overall seismic resistance.

The original cover from the existing column sections was removed, and additional longitudinal reinforcement was added with M20 grade concrete. The ‘concrete02’ material model was used to simulate the jacketed concrete, while Mander’s model [32] was employed to determine the parameters of the confined concrete enclosed by the added longitudinal reinforcement. The ‘steel02’ material model was used to represent the additional longitudinal reinforcements. Figure 6(b) illustrates a typical section of RC jacketing, along with the corresponding material models.

3.2 Steel Jacketing

Retrofitting RC buildings with steel angles and battens is a widely used technique to enhance the strength and deformation capacity of beams and columns in existing structures [45]. This method is also effective for addressing issues related to improper splicing of longitudinal bars [46, 47]. By encasing frame member with steel plates, steel jacketing can significantly improve the seismic resilience of structures. In this study, steel jacketing was implemented by positioning steel angle sections at each corner of the existing

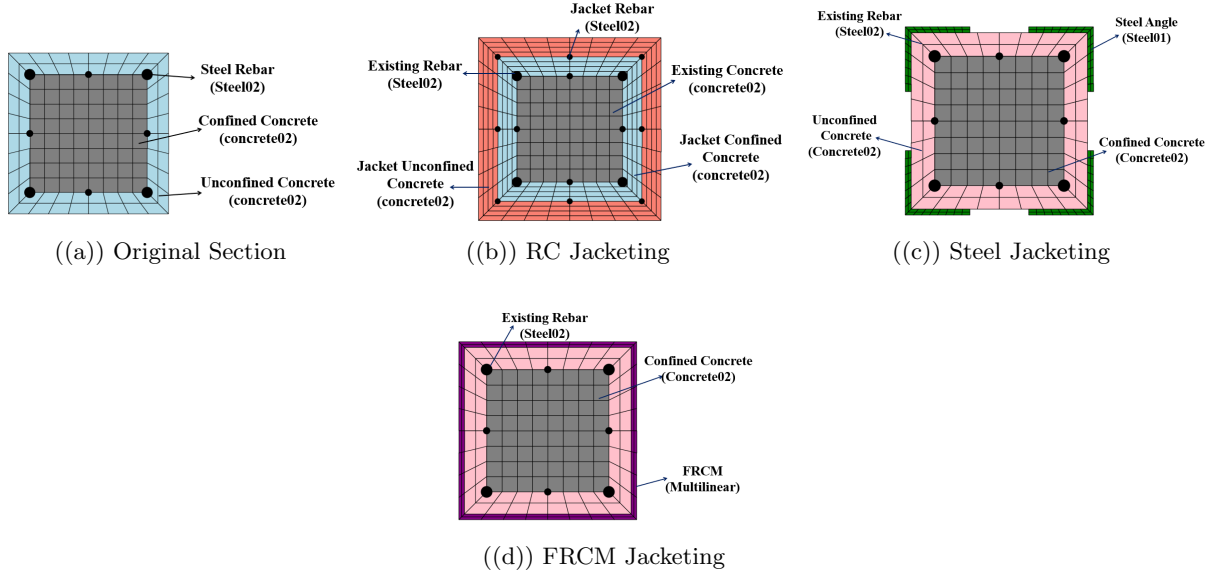


Figure 6: Illustrations of fiber sections for original section and adopted retrofitting methods.

RC columns. The selection of section sizes was based on a trial-and-error approach, observing whether the pushover curve of the deficient non-engineered or pre-engineered building approached that of the engineered building. However, in most cases, the size of steel jacket was governed by the minimum recommended section of a $50 \text{ mm} \times 50 \text{ mm} \times 5 \text{ mm}$ steel angle, as specified in the “Seismic Retrofitting Guidelines of Buildings in Nepal.”

Steel jacketing can be configured to provide either additional confinement alone or both confinement and flexural resistance. In fiber-section modeling of steel jacketing, Campione et al. [48] addressed this distinction. For cases where only confinement is considered, steel angles are excluded from the fiber cross-section. Conversely, in cases where full flexural connection is required, the steel angles are discretized into fibers with specific uniaxial properties. In this study, the second approach was adopted, modeling the steel angles with the ‘Steel01’ uniaxial material in OpenSees to capture flexural effects, without modifications to account for confinement action. Figure 6(c) illustrates a typical section of steel jacketing, along with the corresponding material models.

3.3 Hemp FRCM Jacketing

Fiber Reinforced Cementitious Matrix (FRCM) is a composite material comprising a fiber mesh embedded within a cementitious or lime-based mortar, where the fiber provides tensile reinforcement to the brittle mortar matrix. The tensile properties of the fiber mesh are critical in defining the tensile behaviour and overall structural performance of FRCM systems [21]. Research by Mercedes et. al. [49] has demonstrated that among various vegetal-based FRCM composites, hemp offers the highest mechanical properties, attributed to its favourable strength, ductility, and environmental sustainability. Additional studies by Sapkota et al. [21] have highlighted that hemp-based FRCM retrofitting can significantly improve the structural resilience of non-engineered buildings, especially in seismic zone, by enhancing the lateral load-bearing capacity.

To accurately model the non-linear behaviour of FRCM, a trilinear stress-strain constitutive model was adopted, as illustrated in Figure 7. In the initial elastic phase (Stage I), both the mortar and the fiber mesh deform together, forming the first linear segment in the constitutive model. The second phase (Stage II) begins with the onset of cracking in the mortar matrix, which weakens the bond at the matrix-fiber interface, leading to a partial transfer of stress from the matrix to the fiber mesh. This stage is typically represented by a horizontal segment, assuming a constant stress level as cracks propagate, while the FRCM sustains a residual load-carrying capacity through stress transfer across the cracks. The final phase (Stage III) occurs once the cracking stabilizes, with the fiber mesh assuming the primary load-bearing role, thus resulting in a positive slope in the stress-strain relationship. This stage signifies the full mobilization of the fiber mesh, enhancing the structural integrity of the FRCM element under continued loading [21, 49, 50].

In the present study, a 100 mm thick layer of hemp-based FRCM was applied to the outer surface of

existing structural columns, optimized through trial and error to maximize load-bearing efficiency. This layer was modeled using a trilinear stress-strain curve as characterized by Mercedes et al. [49]. The model implemented the “Multilinear” command in OpenSees to create a multi-linear elastic uniaxial material, capable of representing the trilinear stress-strain behavior of hemp FRCM in tension. Although this response remains nonlinear, it is characterized by elastic behavior across each segment. The mechanical properties used in this FRCM model are summarized in Table 7. Figure 6(d) illustrates a typical section of FRCM jacketing, along with the corresponding material models.

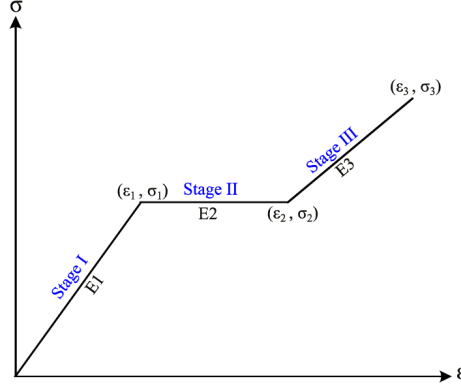


Figure 7: Stress-strain constitutive laws for FRCM Tension [49]

Table 7: Mechanical properties used for FRCM modeling [49]

Parameter of Hemp	Values
Stress at first crack formation, σ_1 (MPa)	4.82
Stress during multi-cracking zone, $\sigma_2 = \sigma_1$ (MPa)	4.82
Ultimate stress of FRCM, σ_3 (MPa)	6.54
Strain at the end of stage II, ε_2 (%)	0.8
Young’s modulus at stage I, E_1 (GPa)	9.32
Young’s modulus at stage III, E_3 (GPa)	0.39

3.4 Glass FRCM Jacketing

Glass FRCM composites exhibits notable mechanical performance under tensile stress, whose behavior is influenced by the properties of their constitutive materials and quality of bond between them. The tensile strength of Glass FRCM systems is primarily governed by the intrinsic strength of glass fiber and the interface performance between the fibers and the inorganic matrix [51, 52]. The application of Glass FRCM jackets improves confined compressive strength and axial strain capacity, while also minimizing strength degradation [53]. In structural modeling, the Glass FRCM’s capacity to sustain tensile loads can be approximated as a tri-linear curve similar to Hemp FRCM. In this study, a 3mm thickness was wrapped was placed around the columns. The stress-strain values for each stage were derived from the average results of tensile test experiments conducted by Carozzi et al. [53], as summarized in Table 8. The corresponding stress-strain behavior of GFRCM is illustrated in Figure 8.

4 Seismic Fragility Analysis

Seismic fragility quantifies the likelihood that a structural component or system will fail to meet acceptable performance levels under seismic forces [54]. Fragility curves illustrate structural vulnerability by correlating various damage states with ground motion intensity measures, such as peak ground acceleration (PGA), peak ground velocity (PGV), or spectral acceleration (Sa) [55]. In this analysis, PGA has been selected as the ground motion intensity parameter.

Incremental Dynamic Analysis (IDA) is a parametric method that involves applying multiple ground motion records to a structural model, with each record scaled across varying intensity levels [56]. The

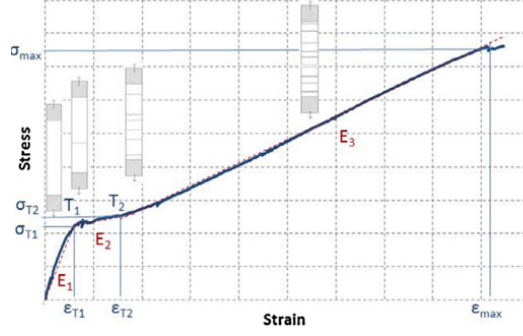


Figure 8: Stress-strain behavior of GFRCM subjected to tensile test [53]

Table 8: Tensile test Results in GFRCM [53]

Parameters of GFRCM	Values
Tensile Stress in the first transition point, σ_{t1} (MPa)	460.05
Tensile Strain in the first transition point, ε_{t1} (%)	0.045
Tensile Stress in the second transition point, σ_{t2} (MPa)	431.11
Tensile Strain in the second transition point, ε_{t2} (%)	0.38
Ultimate Tensile Stress, σ_u (MPa)	872.17
Ultimate Tensile Strain, ε_u (%)	0.69

result is a series of response curves plotted against intensity levels, providing a detailed picture of structural response. IDA has evolved as an extension of non-linear static pushover analysis and has become a standard tool in the seismic evaluation of structures [57, 58]. According to the model described by Vamvatsikos et al. [40], the median seismic demand (Drift) for a given IM can be represented using the following equation, allowing for the determination of seismic demand based on the PSDM.

$$D = a \cdot IM^b \quad (1)$$

Rearranging with the natural logarithm on both sides, we get:

$$\ln(D) = \ln(a) + b \ln(\text{GIM}) \quad (2)$$

where GIM is the ground intensity measure and a , b are regression coefficients. The plot between $\ln(D)$ and $\ln(\text{GIM})$ is called the Probabilistic Seismic Demand Model (PSDM).

Considering the drift limit mentioned in the fragility curve proposed by [59], it is used to develop the fragility curve. The probability of exceedance of limit states for the given ground motion is given by:

$$P\left(\frac{LS_i}{\text{GIM}}\right) = 1 - \phi\left(\frac{\lambda_{CL}^i - \lambda_{D/\text{GIM}}}{\beta_{D/\text{GIM}}}\right) \quad (3)$$

where the parameters of the mean and standard deviation are defined by:

$$\lambda_{D/\text{GIM}} = \ln(a) + b \ln(\text{GIM}) \quad (4)$$

$$\beta_{D/\text{GIM}} = \sqrt{\frac{\sum_{k=1}^n [\ln(\text{GIM}_k) - \lambda_{\text{GIM}}(\text{GIM}_k)]^2}{n - 2}} \quad (5)$$

Where:

n = Number of inter-storey drift data points in IDA plot,

λ_{CL}^i = $\ln$ (average drift capacity for the i^{th} limit state),

GIM = Ground Motion Intensity in g .

4.1 Limit States

In this study, three damage states are defined using the interstory drift ratio (IDR), which serves as an indicator of structural performance. It is important to emphasize that different IDR thresholds have been assigned to various building typologies, reflecting their unique ductility characteristics and structural responses. The limit states employed in deriving fragility curves are obtained from a range of literature sources. The IDR limits for engineered and non-engineered buildings are based on the values proposed by Hemchandra et al. [10]. However, no specific drift limit is established for pre-engineered buildings. As a result, the drift limit for pre-engineered buildings is taken as the average of engineered and non-engineered buildings, given their intermediate performance. For retrofitted buildings, the drift limits are assumed to match those of engineered buildings, as retrofitting aims to reach performance to the level of engineered structures. The IDR limits for different building types are presented in Table 9.

Table 9: Damage limit states drift (%) for different buildings [10]

Level	Engineered	Pre-Engineered	Non-Engineered	Retrofitted
Operational	1	0.85	0.7	1
Life Safety	2.15	1.825	1.5	2.15
Collapse Prevention	3.5	3	2.5	3.5

5 Result and Discussion

5.1 Eigenvalue Analysis

Eigenvalue analysis is used to determine the dynamic properties of structures, including natural mode shapes, time periods and modal participation factors. These properties are dependent on the mass and stiffness of the structure. The fundamental time periods obtained from eigenvalue analysis for non-engineered and pre-engineered buildings are presented in Table 10. Engineered buildings have lower time periods than both original and retrofitted non-engineered and pre-engineered structures, indicating superior stiffness. The lower stiffness in non-engineered and pre-engineered buildings is due to substandard ductile detailing, member sizes and material properties. Retrofitting has reduced the time periods of non-engineered and pre-engineered buildings, bringing them within a range comparable to engineered structures. This suggests that retrofitted buildings have achieved stiffness levels similar to code-compliant engineered typologies. RC jacketing significantly increases stiffness thereby reducing the time period and attracting higher forces due to the increased rigidity.

Table 10: Fundamental time period of all building models

Building ID	Building Type	No. of Storey	Building Typology	Fundamental Time Period (sec)					
				Original	RC Jacket	Steel Jacket	Glass FRM	Hemp FRM	Engineered
Model 1	Non	3	Plan Symmetric	0.93	0.4	0.65	0.53	0.56	0.54
	Pre			0.77	0.43	0.57	0.53	0.52	
Model 2	Non		Plan Irregularity	0.9	0.42	0.59	0.52	0.53	0.48
	Pre			0.76	0.41	0.55	0.48	0.5	
Model 3	Non		Vertical Irregularity	0.87	0.39	0.61	0.5	0.51	0.46
	Pre			0.74	0.38	0.56	0.47	0.48	
Model 4	Non	4	Plan Symmetric	1.17	0.56	0.82	0.57	0.69	0.65
	Pre			1.03	0.57	0.79	0.59	0.74	
Model 5	Non		Plan Assymetric	0.95	0.51	0.63	0.68	0.86	0.85
	Pre			0.93	0.49	0.74	0.67	0.83	
Model 6	Non		Vertical Irregularity	1.35	0.79	1.05	0.8	0.99	0.86
	Pre			1.22	0.79	0.99	0.75	0.92	

Eigenvalue analysis offers valuable insights into the dynamic behavior of structures by identifying natural frequencies and mode shape. However, it has limitations, as it does not account for material degradation and nonlinear behavior. While eigenvalue analysis provides a useful initial approach and helps establish a baseline, it should be integrated with pushover analysis and non-linear time history

analysis. This method addresses full spectrum of dynamic, nonlinear and material-specific challenges in structural retrofitting. A motivation for performing the eigenvalue analysis is to set Rayleigh damping parameters, whose value is used while performing non-linear time history analysis.

5.2 Non-Linear Static Analysis: Pushover Analysis

The non-linear characteristics of the building models were examined using a consistent modeling approach and a monotonic pushover analysis was performed on each building type. Figures 9 and 10 illustrate the pushover curves for non-engineered and pre-engineered buildings respectively. The analysis reveals that the non-engineered building, represented by the red band, exhibits the lowest base shear capacity, followed by the pre-engineered model shown in blue, and finally, the engineered model shown in green, which demonstrates the highest capacity. Each coloured band represents variation in structural typologies within each category by capturing the range of responses for six different typologies.

The pushover curves indicate that the peak base shear values at various drift levels: 283 kN at a 0.96% drift for non-engineered buildings, 377 kN at a 1.2% drift for pre-engineered buildings, and 860 kN at a 1.95% drift for engineered buildings. These results clearly demonstrate that engineered buildings possess significantly greater strength and stiffness compared to their non-engineered and pre-engineered buildings. The higher capacity of the engineered model reflects its compliance with seismic performance standards, as mandated by NBC 105:2020. In contrast, both non-engineered and pre-engineered models fall short of these requirements, indicating the retrofitting is necessary to achieve the desired level of seismic resilience.

To enhance the performance of non-engineered and pre-engineered buildings, seismic retrofitting was implemented using four distinct methods, followed by pushover analysis to assess their effectiveness. The retrofitted models were evaluated for improvements in base shear capacity, strength and stiffness. After retrofitting, the base shear capacity of both building types increased significantly, RC jacketing, for instance, raised the base shear capacity from 283 kN to 810 kN for non-engineered building. For pre-engineered buildings, RC jacketing increased the base shear capacity from 377 kN to 800 kN, corresponding to base shear coefficient of 0.34, which is close to targeted engineered design. Similarly, other retrofitting measures brought both the non-engineered and pre-engineered buildings closer to the performance of the engineered one, which has average base shear of 863 kN and average base shear coefficient of 0.37. Notably, glass FRCM demonstrated a remarkable enhancement in performance, achieving the highest base shear values of 930 kN for non-engineered buildings and 970 kN for pre-engineered buildings. The initial tangential stiffness of glass FRCM surpassed that of the other retrofitting methods. All techniques attained peak base shear values that were nearly equivalent to those of the engineered frame, as detailed in Table 11..

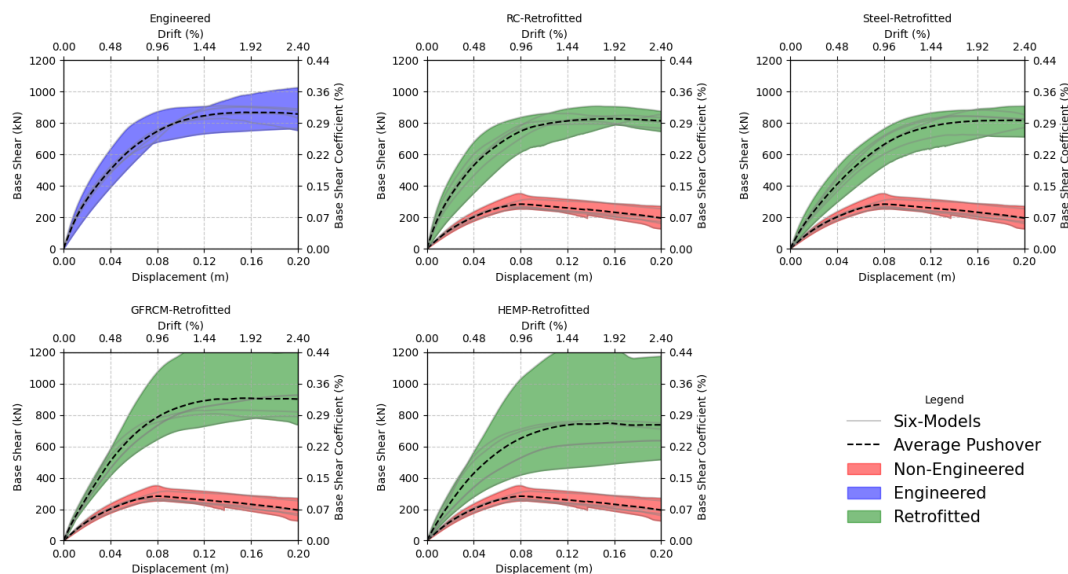


Figure 9: Pushover curves of all building models with comparison of Non-Engineered models with retrofitted counterparts

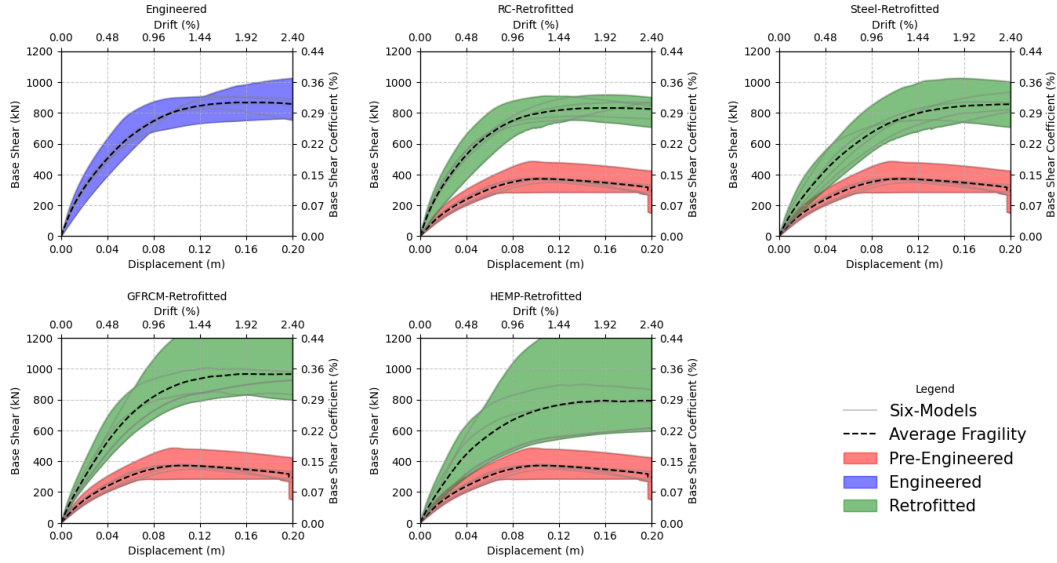


Figure 10: Pushover curves of all building models with comparison of Pre-Engineered models with retrofitted counterparts

Table 11: Average peak base shear and base shear coefficient of all buildings

Retrofitting	Non-engineered		Pre-engineered		Target Engineered Design	
	Base Shear (kN)	Base Shear Coefficient	Base Shear (kN)	Base Shear Coefficient	Base Shear (kN)	Base shear Coefficient
RC Jacket	810	0.36	800	0.34	863	0.37
Steel Jacket	800	0.36	900	0.38		
Glass FRCM Jacket	930	0.41	970	0.41		
Hemp-FRCM Jacket	748	0.33	776	0.33		

5.3 IDA curves

The IDA curves for Model 3, shown in Figure 11, illustrate the seismic performance of non-engineered, pre-engineered, and engineered buildings under incremental loading up to a peak ground acceleration (PGA) of 0.5g, with 0.1g increments. Please note, similar curves for other building models are not presented to maintain brevity. As depicted in Figure 11(a), the non-engineered building surpasses the LS limit state (1.5% drift) at 0.1g PGA, while the pre-engineered building reaches its LS limit state (1.825% drift) at approximately 0.17g PGA. In contrast, the engineered building attain its LS limit state (2.15% drift) around 0.35g PGA. These results highlight the vulnerability of non-engineered and pre-engineered buildings to even low-intensity seismic events, whereas the engineered building produces superior performance. After retrofitting the building, it is observed that the probability of exceeding the limit state has decreased. For instance, Figure 11(d) shows the IDA curve for Glass-FRCM jacketing. The LS limit state (1.5% drift) is exceeded at 0.3g and 0.35g for retrofitted non-engineered and retrofitted pre-engineered buildings, respectively. These IDA curves are utilized in subsequent section for fragility analysis.

5.4 Fragility analysis

5.4.1 Non-engineered Buildings

The fragility curves for non-engineered buildings across various performance states—Immediate Occupancy (IO), Life Safety (LS), and Collapse Prevention (CP) are depicted in Figures 12, 13, and 14. At the IO state, as shown in Figure 12, non-engineered buildings reached a 100% probability of exceedance

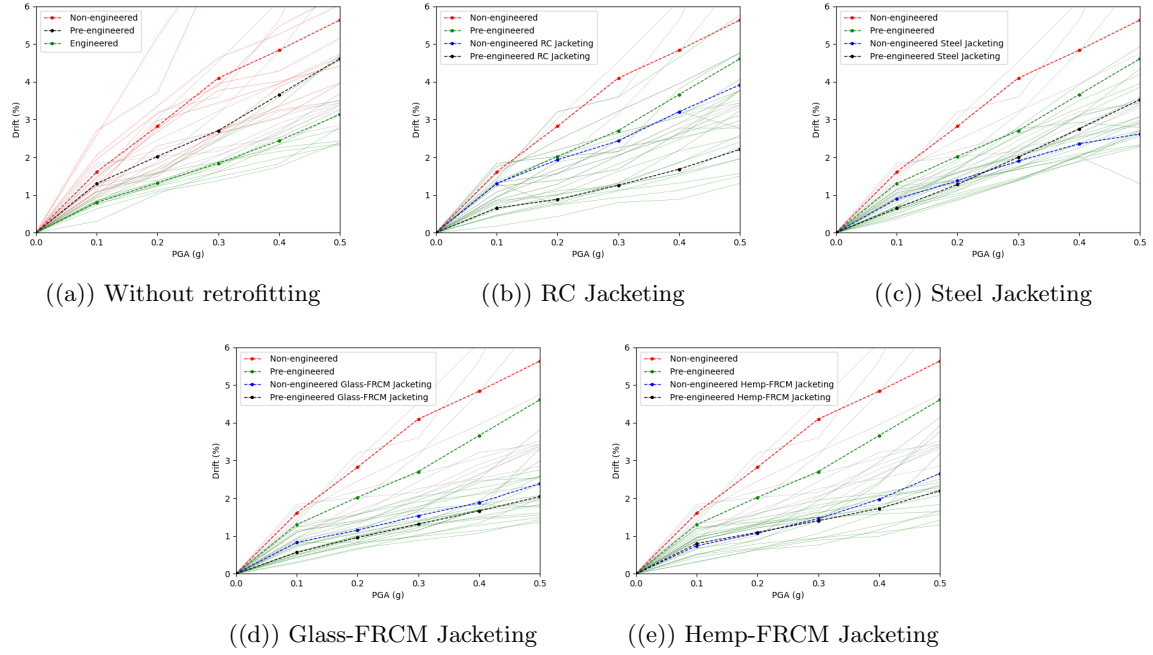


Figure 11: Comparison of IDA Curves of Model 3

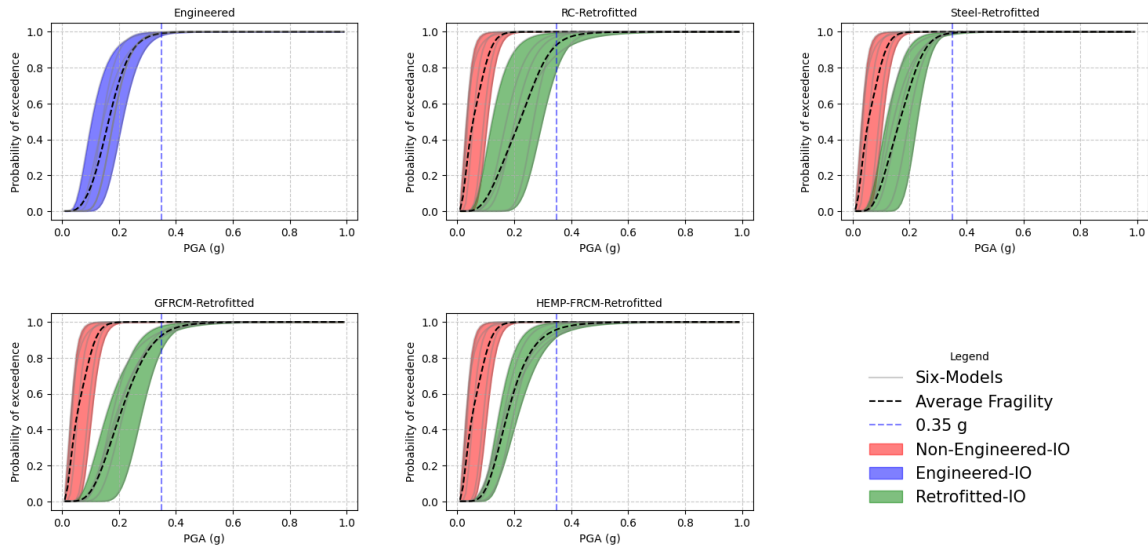


Figure 12: Comparison of fragility of non-engineered building with engineered and retrofitted counterparts in IO limit state

at a low PGA (below 0.2g), represented by the red band, while engineered buildings achieved the same threshold only around 0.4g. Retrofitting with RC Jacketing, particularly with a 100 mm thickness, demonstrated superior effectiveness, achieving a 90% probability of IO exceedance at 0.35g, outperforming even engineered buildings. Similarly, at the LS state (Figure 13), non-engineered buildings reached 100% exceedance at 0.35g, compared to only 40% for engineered models. Among retrofitting techniques, RC Jacketing and glass-FRCM reduced LS exceedance probabilities to under 15%, surpassing engineered performance. At the CP state (Figure 14), non-engineered buildings faced a 90% probability of collapse at 0.35g, while engineered structures, compliant with NBC 105:2020, reduced this risk to below 10%. RC Jacketing, steel jacketing, and FRCM variants significantly enhanced collapse resistance. The findings indicate that non-engineered buildings are especially vulnerable at high PGA levels, emphasizing the need for robust retrofitting methods such as RC Jacketing and GFRM to substantially improve

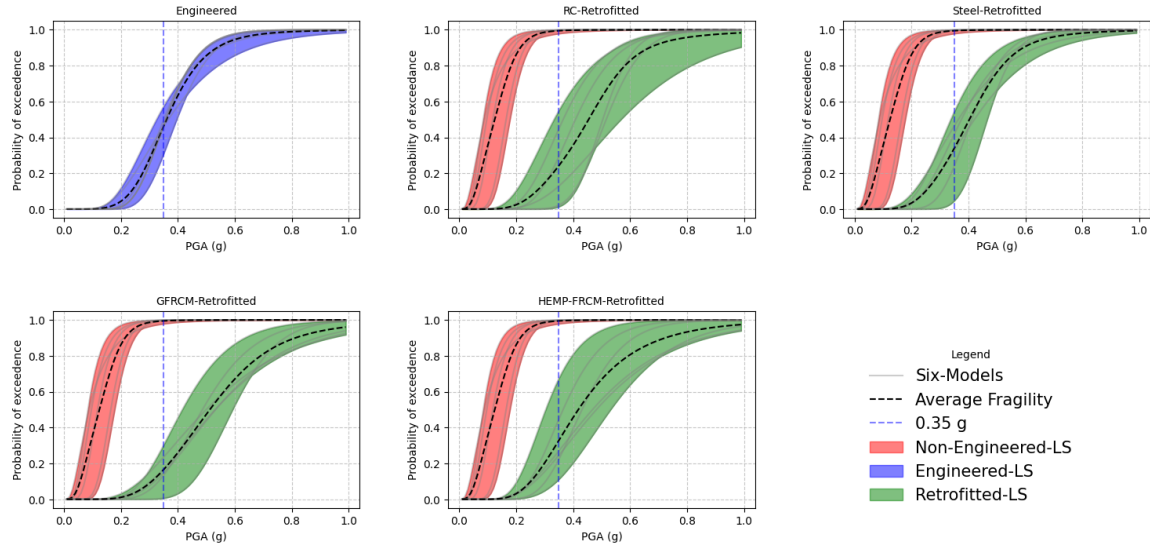


Figure 13: Comparison of fragility of non-engineered building with engineered and retrofitted counterparts in LS limit state

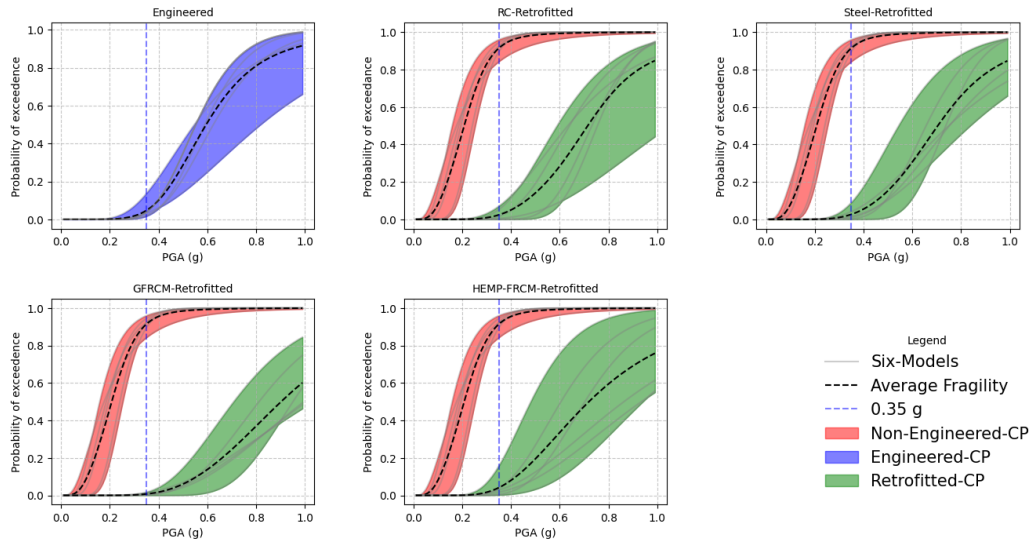


Figure 14: Comparison of fragility of non-engineered building with engineered and retrofitted counterparts in CP limit state

resilience.

5.4.2 Pre-engineered Buildings

The fragility curves of pre-engineered buildings across various performance states - IO, LS and CP - are displayed in Figures 15, 16 and 17. Pre-engineered buildings demonstrate a significant increase in resilience compared to Non-Engineered structures. As shown in Figure 15, these buildings are more likely to reach the IO state, even at PGA levels, with a 60% chance of remaining within the IO state under 0.35g PGA. Both RC and steel jacketing significantly enhance performance, while glass-FRCM and hemp FRCM are effective to a lesser degree. For the LS state, Figure 16 shows that pre-engineered buildings also have a reduced probability of exceedance compared to non-engineered buildings, ranging from 85% to 90% at a PGA of 0.35g. Among retrofitting options, GFRCM proves most effective in reducing LS exceedance, while RC and Steel Jacketing achieve a similar enhancement, reducing exceedance probabilities to below

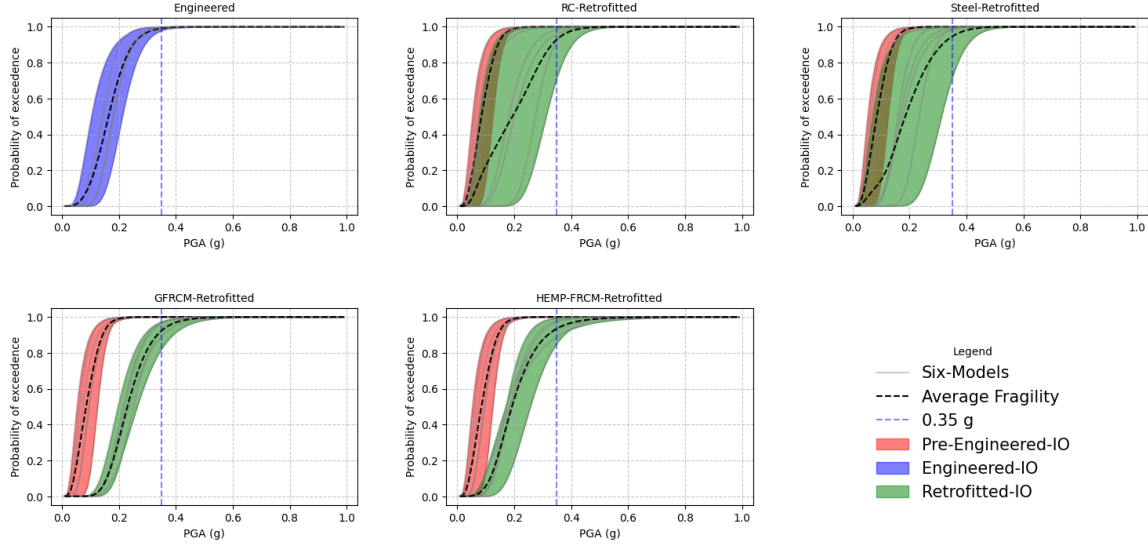


Figure 15: Comparison of fragility of pre-engineered building with engineered and retrofitted counterparts in IO limit state

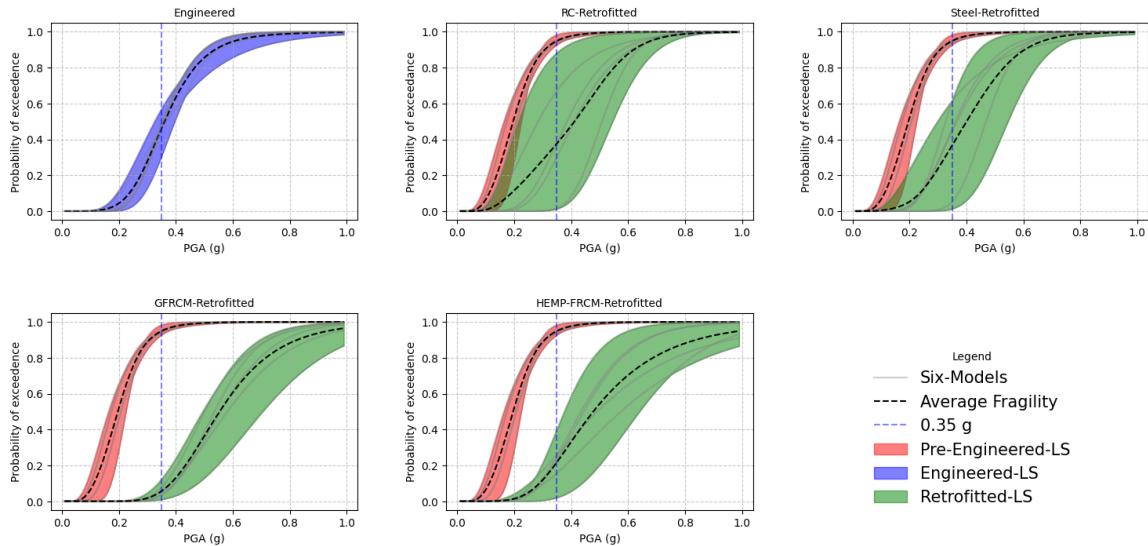


Figure 16: Comparison of fragility of pre-engineered building with engineered and retrofitted counterparts in LS limit state

35%. At the CP state, as depicted in Figure 17, Pre-Engineered buildings show improved collapse resistance, with approximately 80% probability of failure under high PGA values, slightly better than Non-Engineered buildings. However, GFRM and Hemp FRCM wraps provide a marked improvement, reducing collapse probabilities to less than 5%. RC and Steel Jacketing with minimal thicknesses bring these structures close to code-compliant standards.

5.5 Discussion

The dynamic analysis highlights the significant discrepancies in stiffness and seismic performance between non-engineered, pre-engineered, and engineered building typologies. Eigenvalue analysis reveals that engineered buildings designed in compliance with NBC 105:2020 exhibit substantially lower fundamental time periods, indicating greater stiffness and adherence to seismic resilience standards in comparison to

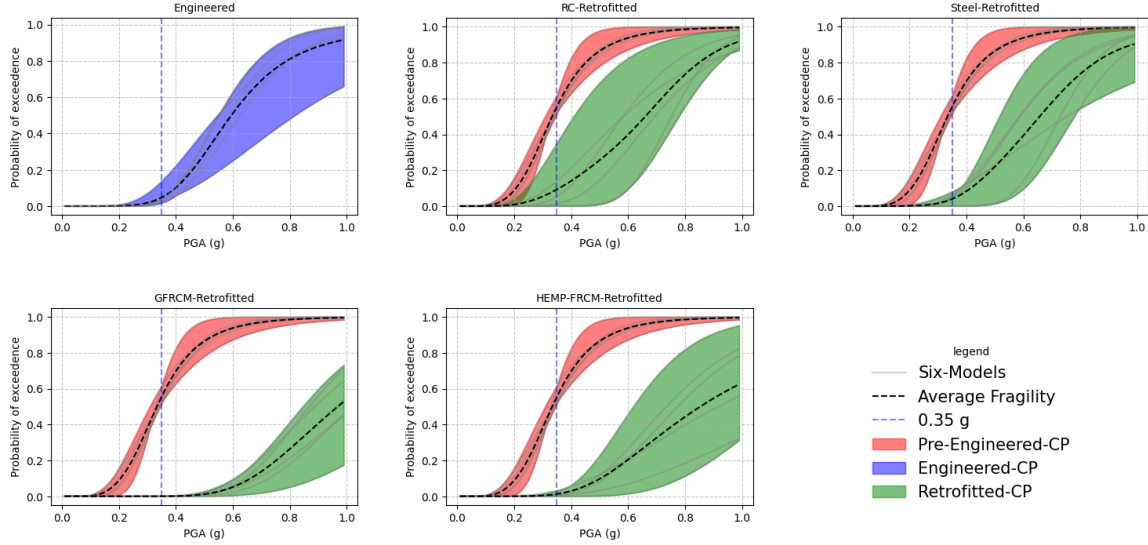


Figure 17: Comparison of fragility of pre-engineered building with engineered and retrofitted counterparts in CP limit state

non-engineered and pre-engineered counterparts. The effectiveness of various retrofitting techniques was analyzed through both eigenvalue and non-linear pushover analysis. Retrofitting, especially through RC Jacketing and glass-FRCM, successfully reduced the time periods of non-engineered and pre-engineered buildings, thus bringing their dynamic behavior closer to that of engineered structures.

Pushover analysis further validates the improvements in seismic performance post-retrofitting, with all methods, significantly enhancing base shear capacities and drift levels. RC Jacketing, steel jacketing, and GFRCM each raised the base shear of non-engineered structures to near-engineered levels, while also extending allowable drift, which implies an increased capacity to withstand seismic deformations. The fragility curves provide a comparative perspective on retrofitting efficacy across performance states. At IO, non-engineered buildings reached high probabilities of exceedence even at low PGA levels (below 0.2g), while retrofitted buildings, especially with RC Jacketing, demonstrated a marked improvement, achieving resilience levels exceeding those of engineered models. In the LS and CP states, retrofitted buildings with RC Jacketing and GFRCM exhibited the lowest exceedence probabilities, effectively enhancing resilience against potential collapse under significant seismic demands.

Overall, these analyses illustrate that robust retrofitting strategies can significantly bridge the performance gap between non-engineered/pre-engineered and engineered structures. Integrating eigenvalue analysis with pushover and fragility analysis provides a comprehensive understanding of retrofitted structures' dynamic and non-linear response, highlighting that advanced retrofitting techniques such as RC and steel jacketing are crucial for enhancing the seismic resilience of non-engineered buildings. This multi-method approach underlines the importance of selecting retrofitting techniques tailored to achieve structural compliance and ensure life safety, especially in high seismic zones where non-engineered buildings are predominant.

6 Conclusion

The study examines the seismic vulnerability of six existing non-engineered and pre-engineered buildings and evaluates the effectiveness of six retrofitting interventions. Finite element models were developed in OpenSees to perform eigenvalue analysis, pushover analysis, and dynamic time history analysis, with fragility curves generated through incremental dynamic analysis. The results demonstrate that retrofitting significantly enhances the strength, stiffness, and ductility of the buildings. However, the effectiveness of the retrofitting interventions varies notably between cases. For pre-engineered buildings, the impact of retrofitting is comparatively limited, likely due to the larger size of structural members. In contrast, the non-engineered buildings exhibit a greater need for retrofitting to achieve compliance with seismic codes and mitigate risks from future earthquakes. The findings emphasize the urgent need

for retrofitting existing buildings to improve their seismic performance and ensure code compliance. The main conclusions drawn from the analyses are summarized as follows:

- The mean peak base shear capacities of building models of non-engineered, pre-engineered, and engineered buildings are 250 kN, 400 kN, and 900 kN, respectively, with corresponding base shear coefficients of 0.1, 0.17, and 0.4, highlighting the greater vulnerability of non-engineered buildings compared to pre-engineered ones.
- From the pushover analysis, it was found that after retrofitting, all techniques, achieved performance levels comparable to engineered buildings in both non-engineered and pre-engineered cases.
- Fragility analysis revealed that non-engineered and pre-engineered buildings exceeded 100% IO limit state at 0.2g, but after retrofitting, this improved to 0.4g for most interventions, matching the performance of engineered buildings.
- At the LS limit state, probabilities of exceedance 100% and 90% occur at 0.35g for non-engineered and pre-engineered buildings, respectively. After retrofitting, the probability drops to an average of 30% at 0.35g, outperforming even engineered buildings, which have a 50% probability at the same level.
- At the CP limit state, non-engineered buildings have a 90% probability of exceedance at 0.35g, while pre-engineered buildings have a 60% probability. After retrofitting, the probability drops to an average of 10% at 0.35g, closely matching the engineered level.

The study, while comprehensive, has certain limitations. It focuses solely on the Lalitpur Metropolitan City (LMC) and six typologies of low-rise RC buildings, which may limit its generalization to other regions or building types. The retrofitting methods analyzed emphasize local interventions, without exploring global techniques or evaluating the combination of multiple retrofitting strategies. Moreover, the study does not include an economic analysis, leaving the cost-effectiveness of retrofitting measures across diverse socio-economic contexts unexplored. These limitations suggest areas for future research to enhance the applicability and robustness of the findings.

Author contributions

Santosh Aryal: Writing – original draft, Visualization, Investigation, Software analysis; **Smriti Dahal:** Writing – original draft, Investigation, Software analysis; **Suraj Thapa:** Visualization, Investigation, Software analysis; **Sujal Koirala:** Investigation, Software analysis; **Suniti Shrestha:** Writing – original draft, Investigation, Software analysis; **Sushil Raj Pandey:** Software analysis; **Kshitij C. Shrestha:** Conceptualization, Methodology, Visualization, Investigation, Writing - Review and editing, Supervision

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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